
Education

PhD in Aeronautics and Astronautics (Ongoing)

2024 -

Massachusetts Institute of Technology

- Major: Plasma and Lightning Physics | Minor: Data Visualization
- MIT Outing Club Vice President, Graduate Christian Fellowship Treasurer, MIT Veritas Forum lead

MEng Aeronautical Engineering with a Year Abroad (1st class honors)

2020-2024

Imperial College London, Massachusetts Institute of Technology (final year)

- Overall grade: 80% (GPA 5.0). Awarded the BAE Systems Prize for the best individual research project

A-Levels in Maths, Further Maths, Physics, Product Design (A*, A*, A*, A*)

2018-2020

Peter Symonds' College

- Authored a grade A* Extended Project Qualification on ion propulsion for air and space transport

Experience

Graduate Researcher, Electromagnetic Effects

May 2026 - Aug 2026

The Boeing Company, Engineering & Technology Innovation

- Created a new model-based systems engineering tool for aircraft lightning protection
- Designed, executed, and evaluated an experimental campaign to evaluate novel material properties
- Delivered tools and results to stakeholders across the enterprise

Graduate Student, Aeronautics and Astronautics

Sep 2024 -

Massachusetts Institute of Technology

- Developed physics-based simulation tools for aircraft lightning protection
- Collaborated with industry partners through conferences and international standards committees
- Drove model validation using flight test data and high-fidelity computational electromagnetic aerodynamics

Graduate Resident Advisor

Sep 2024 -

Massachusetts Institute of Technology, New Vassar

Student Partner

Sep 2024 -

Giant Ventures

- Identified and referred exceptional ventures to a global firm focused on technology, health, and climate

Summer Intern, Simulation & Modeling

Jun 2022 - Aug 2022

MBDA Missile Systems

- Evaluated and implemented an alternative programming paradigm in a high-fidelity dynamic model
- Identified a solution to a long-term project which protected company IP while meeting client requirements

Undergraduate Researcher (UROP)

Jun 2021 - Aug 2021

Imperial College London, Dr Sylvain Laizet (Supervisor)

- Implemented and profiled a novel C++ framework for high-performance heterogeneous computing

Duty Manager

Mar 2020 - Jan 2021

The Food Warehouse, Basingstoke

Lead Aerodynamics and Simulations Engineer, Altitude Record Team

Oct 2020 - Aug 2023

Imperial College London Rocketry

- Conducted computational fluid dynamics (CFD) studies on high-powered supersonic rockets using StarCCM+
- Negotiated sponsorship with ESTECO, acquiring licenses to modeFrontier optimization software
- Integrated modeFrontier into engineering workflows, increasing rocket altitude by 3% and breaking a UK record

F1inSchools (STEM Racing) Alumnus, Judge, and Event Volunteer

Sep 2017 - Jul 2021

Publications & Presentations

An End-to-End Simulation-Based Workflow for Zoning

Oct 2026

  International Conference on Lightning and Static Electricity (ICOLSE) | Charleston, SC

Aircraft Lightning Protection: Zoning by Simulation

May 2026

 Future Leaders in Aerospace Symposium | Boulder, CO

Aircraft-Scale Modeling of Lightning Phenomena: Zoning by Simulation

May 2026

 MIT-Boeing Annual Review | Massachusetts Institute of Technology

A Physics-Based Approach to Aircraft Lightning Zoning: Zone 2

Oct 2025

 IEEE Access, vol. 13, pp. 188564-188576

Recent Progress in Zoning by Simulation

Oct 2025

 SAE AE-2 Committee Meeting | Pittsburgh, PA

Mapping of Swept Stroke Physical-Quantities to Zoning Diagrams

Jul 2025

  EUROCAE WG 31 Zoning Workshop | Paris, France

A Numerical Model for the Zoning of Unconventional Aircraft: The Swept Stroke Phase

Apr 2025

 MIT-Boeing Annual Review | Massachusetts Institute of Technology

Numerical Simulation of the Lightning Swept Stroke:

Dec 2024

Application to the Results from the NASA Storm Hazards Program

 IEEE Access, vol. 12, pp. 188231-188244

Development and Validation of Numerical Models for Lightning Zoning Assessment

Oct 2024

 SAE AE-2 Committee Meeting | Cape Canaveral, FL

Physics-Based Zoning of Unconventional Aircraft: The Swept Stroke Phase

Sep 2024

  International Conference on Lightning and Static Electricity (ICOLSE) | Campinas, Brazil

Numerical Simulation of the Lightning Swept Stroke for the Zoning of Unconventional Aircraft

Jun 2024

 Masters Thesis | Imperial College London

The Lightning Swept Stroke: A Numerical Model for the Zoning of Unconventional Aircraft

Apr 2024

 MIT-Boeing Annual Review | Massachusetts Institute of Technology

Awards

Mathworks Engineering Fellowship

2025, 2026

Mathworks Inc.

Graduate Fellowship

2024

MIT School of Engineering

Engineering Simulation Techniques: Outstanding Undergraduate Project Award (2nd prize)

2024

NAFEMS and the Institution of Mechanical Engineers

BAE Systems Prize

2024

Imperial College Dept. of Aeronautics, best individual research project

Aeronautics Scholar Imperial College Dept. of Aeronautics	2022
Dean's List Imperial College Dept. of Aeronautics, top 10% of cohort	2021, 2022, 2023, 2024
James Clayton Undergraduate Scholarship Institution of Mechanical Engineers	2020 - 2024
Arkwright Scholar The Smallpeice Trust	2018